

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)			DMFT award (1) for any of following - the higher the fluoride concentration, the lower the mean DMFT / the lower the fluoride concentration, the higher the mean DMFT / OWTTE - at this concentration DMFT is low - at lower concentration DMFT is high / higher award (1) for increasing fluoride ion concentrations above 1.0 mg/dm³ does not make a difference to DMFT levels / OWTTE fluorosis award (1) each for any two of following - the higher the fluoride concentration, the higher the percentage affected by fluorosis / the lower the fluoride concentration, the lower the percentage affected by fluorosis / OWTTE - at this concentration fluorosis is low - at higher concentration fluorosis increases <u>significantly</u> / when fluoride ion concentration reaches 1.2 mg/dm³ there is a <u>significant</u> increase in fluorosis (to 35%) / OWTTE - this concentration is best/optimum balance between DMFT and fluorosis			4	4		
	(b)			chlorine kills bacteria / sterilises drinking water / makes the water safe to drink (1) award (1) for either of following - fluoride can cause side effects e.g. can cause stomach cancer, bone cancer, birth defects, infertility, brittle bones, IBS - adding fluoride is a form of mass medication / people are forced to consume it	2			2		

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					AO1	AO2	AO3	Total	Maths	Prac
	(c)			titanium fluorine $\frac{2.4}{48}$ $\frac{3.8}{19}$ (1) 0.05 0.2 (1) TiF ₄ (1) award (2) for Ti ₄ F with working shown ecf possible		3		3	3	
				Question 6 total	2	3	4	9	3	0